

Ribbon operators that add a single shape and still commute the structure of the vanishing locus (gap (H2) of Q150)

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16 September 2026 — prove cycle 2

Abstract

The morning session [2] refuted the nonvanishing lemma of [1] and replaced the classification of commuting pairs $[R_e^W, R_f^{\bar{W}}] = 0$ by a commuting-words theorem, reducing everything to a single rank- d weight τ , $d = \gcd(e, f)$, subject to the two-bead condition $(2B_d^*)$. It left open (H2): *which* τ with zeros satisfy it.

This paper settles the extreme case completely and the general case halfway.

(A) For a weight supported on a *single* word $w \in \{0, 1\}^{d-1}$ — an operator that adds exactly one ribbon shape and nothing else — $(2B_d^*)$ holds if and only if

$$w_{p+1} \neq w_{n-p} \quad \text{for every border length } p \text{ of } w, \quad n = d - 1.$$

The border $p = 0$ alone reads: *exactly one of the two end rows of the ribbon has length 1*. Consequently, for every $d \geq 3$, every such w and every $k \geq 2$, the operator $R_d^{k^w}$ — which adds the single ribbon shape $\alpha(w) \models d$ — commutes with an explicit $R_{kd}^{\bar{W}}$ supported on 2^{k-1} shapes. There are 2, 4, 8, 12, 28, 48, 104 such shapes for $d = 3, \dots, 9$. This strictly sharpens [2, Thm. 6.6], whose weights are supported on 2^{d-3} shapes.

(B) For the strong condition $(2B_d^{**})$ the single-shape solutions are exactly the *antipalindromes* $w_i \neq w_{n+1-i}$; in particular there are none for even d .

(C) In general: every support of a $(2B_d^*)$ -solution is empty, everything, or contained in one of the cylinders $\mathcal{C}_{j,b}$ of [2, Conv. 6.4] — indeed the cylinders are exactly the maximal proper supports — and for the minimal such j the support is invariant under flipping every coordinate in $\{1, \dots, j-1\} \cup \{d-j+1, \dots, d-1\}$. This proves the containment half of (H2), which [2] had only observed computationally for $d \leq 5$. Two things follow. It explains the threshold $\gcd \geq 3$: $d \geq 3$ is precisely when $[1, d-1]$ contains an index $j \neq d/2$. And it turns the refuted lemma into a genuine *dichotomy*: a commuting W either vanishes nowhere, or vanishes at more than three quarters of the ribbon shapes of rank d — at all but a fraction $4^{-e/d}$ of those of rank e . There is nothing in between.

Independently of all this, the (3, 6) counterexample of [2, Thm. 6.7] is re-verified here from the operator definition by a from-scratch implementation, and one omission in its statement is corrected: the value $W_{(1,2)}$, which that theorem does not specify, must be 0.

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1 What this paper does, and what it does not

Let λ be a partition, $M(\lambda)$ its Maya set, and for a *shape weight* $W : \{0, 1\}^{e-1} \rightarrow K$ let R_e^W be the operator of [1, Conv. 1.2],

$$R_e^W |M\rangle = \sum_{b \in M, b+e \notin M} W(u_M(b, e)) |(M \setminus \{b\}) \cup \{b+e\}\rangle, \quad u_M(b, e)_i = [b+i \in M].$$

By [1, Lemma 1.3] the word u names the shape of the added ribbon: $u \mapsto \alpha(u)$ is a bijection onto compositions of e , the parts being the row lengths read top row first.

[2] proved that $[R_e^W, R_f^{\bar{W}}] = 0$ with $W \not\equiv 0 \not\equiv \bar{W}$ forces, in the sign-free coordinates $\sigma(u) = (-1)^{|u|} W(u)$,

$$\sigma = c \tau^{\otimes(e/d)}, \quad \bar{\sigma} = \bar{c} \tau^{\otimes(f/d)}, \quad d = \gcd(e, f),$$

for a single rank- d weight τ , and that the whole remaining content is the two-bead condition ($2B_e^*$) on σ . Question (H2) of [2, §8] is: what are the rank- d solutions with zeros?

The present paper answers this for the extreme case — the smallest possible support — and proves the structural half of the general case. What it does *not* do is give a closed description of the admissible supports *inside* a cylinder $\mathcal{C}_{j,b}$; §10 states precisely what remains, with the reduction that Theorem 6.1 performs.

A remark on why the single-shape case is worth isolating. The refuted lemma [1, Lemma 8.2] said that a commuting partner forces *every* weight value to be nonzero. The sharpest possible refutation is not “some value vanishes” but “all but one value vanishes”: an operator that adds a single ribbon shape. Theorem 4.2 says exactly which single shapes do this, and the answer is a condition on the word’s borders — again a free-monoid statement, as was the gcd in [2, Thm. 1.2].

2 Setting

We use [2, §2] verbatim and do not restate its proofs. Throughout K is an integral domain, $d \geq 1$, and $n := d - 1$.

Convention 2.1 ([2, Conv. 2.1]). A shape weight σ of rank $m \geq 2$ satisfies $(2B_m^*)$ if for every δ with $1 \leq \delta \leq m - 1$ and all $\pi, \varsigma \in \{0, 1\}^{\delta-1}$, $\pi' \in \{0, 1\}^{m-\delta-1}$,

$$\sigma(\pi 0 \pi') \sigma(\pi' 0 \varsigma) = \sigma(\pi 1 \pi') \sigma(\pi' 1 \varsigma), \quad (1)$$

the marked letter sitting at position δ in the first factor and at position $m - \delta$ in the second. It satisfies the stronger $(2\mathbf{B}_m^{**})$ if the same holds with the second factor's blocks decoupled: for all $\alpha, \beta' \in \{0, 1\}^{\delta-1}$ and $\alpha', \beta \in \{0, 1\}^{m-\delta-1}$,

$$\sigma(\alpha 0 \alpha') \sigma(\beta 0 \beta') = \sigma(\alpha 1 \alpha') \sigma(\beta 1 \beta'). \quad (2)$$

For $m = 1$ both are vacuous.

Two facts from [2] are used as stated.

- [2, Thm. 1.3] (the classification): for $1 \leq e < f$, $d = \gcd(e, f)$, over a field F , $[R_e^W, R_f^{\bar{W}}] = 0$ with both weights nonzero *iff* there are a nonzero rank- d weight τ and $c, \bar{c} \in F^\times$ with $\sigma = c\tau^{\otimes(e/d)}$, $\bar{\sigma} = \bar{c}\tau^{\otimes(f/d)}$ and σ satisfies $(2\mathbf{B}_e^*)$.
- [2, Lemma 5.1] (descent), part (a) in the case $k = 1$: if $e = d$ then $\sigma = \tau$, and $(2\mathbf{B}_d^*)$ for τ is exactly $(2\mathbf{B}_e^*)$ for σ .

Definition 2.2 (support condition). For $Z \subseteq \{0, 1\}^n$ say that Z is **admissible** (for rank $d = n + 1$) if for every $\delta \in [1, d - 1]$ and all $\pi, \varsigma \in \{0, 1\}^{\delta-1}$, $\pi' \in \{0, 1\}^{d-\delta-1}$,

$$[\pi 0 \pi' \in Z \wedge \pi' 0 \varsigma \in Z] \iff [\pi 1 \pi' \in Z \wedge \pi' 1 \varsigma \in Z]. \quad (3)$$

Proposition 2.3. *Let σ be a shape weight of rank d over an integral domain and $Z = \{u : \sigma(u) \neq 0\}$. If σ satisfies $(2\mathbf{B}_d^*)$ then Z is admissible. Conversely, if Z is admissible then the indicator $\mathbb{1}_Z$ satisfies $(2\mathbf{B}_d^*)$. Hence the admissible subsets of $\{0, 1\}^n$ are exactly the supports of $(2\mathbf{B}_d^*)$ -solutions.*

Proof. In a domain a product of two elements is 0 iff a factor is 0, so $\sigma(\pi 0 \pi') \sigma(\pi' 0 \varsigma) \neq 0$ iff both arguments lie in Z ; equality of the two sides of (1) therefore forces (3). Conversely for $\sigma = \mathbb{1}_Z$ each side of (1) is 1 if both its arguments lie in Z and 0 otherwise, and (3) says these two alternatives agree. $\square$

Proposition 2.3 is what makes the support question self-contained: no separate existence argument for the values is needed, and every admissible Z really occurs.

3 Independent verification of the (3,6) counterexample, and one correction

Before building on [2] I re-derived its load-bearing object from [1, Conv. 1.2] alone, in a fresh implementation that shares no code with the Q149 or Q150 engines (`code/2026-09-16-c2-Q150-H2/independent_maya`). Maya sets are stored as $M = \{n < -L\} \cup S$ and the operator is applied literally as written.

Calibration. The dictionary [1, Lemma 1.3] was re-checked against Young diagrams: for every partition λ with $|\lambda| \leq 8$, every $e \leq 5$ and every legal bead move, the row lengths of μ/λ were compared with $\alpha(u_M(b, e))$. All 1375 moves agree (the same population size as [1, Check V1], reached independently).

The counterexample. With W and $\bar{W}$ of [2, Thm. 6.7], $[R_3^W, R_6^{\bar{W}}]$ annihilates every one of the 195 partitions with $|\lambda| \leq 11$, and both operators act nontrivially (R_3^W on 21 partitions of size ≤ 6 , $R_6^{\bar{W}}$ on 17 of size ≤ 8). Seven perturbations were run as negative controls; all fire. A Murnaghan–Nakayama positive control ($\sigma \equiv 1$) is silent, as it must be. Details in §9, V1–V2.

A correction to [2, Thm. 6.7]. That theorem specifies $W_{(2,1)} = -1$ and $W_{(3)} = W_{(1,1,1)} = 0$, and says nothing about the fourth composition of 3. The value is not free: control NC1 sets $W_{(1,2)} = 1$ and the commutator becomes nonzero on 35 of the 97 partitions with $|\lambda| \leq 9$. The intended weight is of course $\sigma = \mathbb{K}_{(1,0)}$, which gives $W_{(1,2)} = 0$ — as the abstract of [2] says in words (“adds *only* the L-tromino $(2,1)$ ”). The statement should read

$$W_{(2,1)} = -1, \quad W_{(3)} = W_{(1,2)} = W_{(1,1,1)} = 0.$$

Nothing else in [2] is affected: its proof constructs τ as the indicator of $\mathcal{C}_{1,1} = \{(1,0)\}$ and therefore has the right object throughout. Only the displayed statement is incomplete.

4 Single-shape weights

Definition 4.1. Let $w \in \{0,1\}^n$. An integer p with $0 \leq p \leq n-1$ is a **border length** of w if $w_{[1,p]} = w_{[n-p+1,n]}$. (Thus $p = 0$ is always a border length.) Say w is **non-overlapping** if

$$w_{p+1} \neq w_{n-p} \quad \text{for every border length } p \text{ of } w. \quad (\text{NB})$$

In words: for each way in which a prefix of w coincides with the suffix of the same length, the letter immediately *after* the prefix occurrence and the letter immediately *before* the suffix occurrence must differ.

Theorem 4.2 (single-shape weights, weak condition). *Let $d \geq 2$, $n = d-1$, let K be an integral domain, $c \in K \setminus \{0\}$ and $w \in \{0,1\}^n$. Then $\tau = c\mathbb{K}_w$ satisfies $(2B_d^*)$ if and only if w is non-overlapping.*

Proof. Fix $\delta \in [1, d-1]$ and $\gamma \in \{0,1\}$, and consider the side $\tau(\pi\gamma\pi')\tau(\pi'\gamma\varsigma)$ of (1), where $|\pi| = |\varsigma| = \delta-1$ and $|\pi'| = d-\delta-1 = n-\delta$. Since τ is supported on the single word w , this side is nonzero iff both arguments equal w , i.e. iff

$$\pi = w_{[1,\delta-1]}, \quad \gamma = w_\delta, \quad \pi' = w_{[\delta+1,n]} \quad \text{and} \quad \pi' = w_{[1,n-\delta]}, \quad \gamma = w_{n-\delta+1}, \quad \varsigma = w_{[n-\delta+2,n]},$$

the two prescriptions of π' having the same length $n-\delta$. So, for suitable π, π', ς , the side is nonzero exactly when

$$w_{[\delta+1,n]} = w_{[1,n-\delta]} \quad \text{and} \quad \gamma = w_\delta = w_{n-\delta+1}, \quad (4)$$

and then its value is c^2 .

Substitute $p := n-\delta$, so that p runs over $[0, n-1]$ as δ runs over $[1, n] = [1, d-1]$. The first condition in (4) reads $w_{[n-p+1,n]} = w_{[1,p]}$, i.e. p is a border length of w ; the second reads $\gamma = w_{n-p} = w_{p+1}$.

Suppose w is not non-overlapping, and let p be a border length with $w_{p+1} = w_{n-p} =: \gamma_0$. Take $\delta = n-p$ and the arguments prescribed above. Then the γ_0 -side of (1) equals c^2 , while the $(1-\gamma_0)$ -side is 0 because $\gamma = 1-\gamma_0$ violates the second condition of (4). As K is a domain and $c \neq 0$ we have $c^2 \neq 0$, so (1) fails. (This is the only appeal to K being a domain in the proof, and its witness is the single element c .)

Conversely, suppose w is non-overlapping. Then for every δ and every γ the conditions (4) fail, so both sides of (1) vanish identically and τ satisfies $(2B_d^*)$. $\square$

Remark 4.3 (reading (NB) on the ribbon). The border $p = 0$ contributes the condition $w_1 \neq w_n$. By [1, Lemma 1.3], $w_1 = 1$ iff the *last* part of $\alpha(w)$ is 1 and $w_n = 1$ iff the *first* part is 1. So the $p = 0$ clause says: *exactly one of the two end rows of the ribbon has length 1*. For $d \leq 5$ this clause is the whole of (NB); the first w satisfying it but failing (NB) occurs at $d = 6$, namely $w = 01001$ (border $p = 2$, with $w_3 = w_3$), which has period $3 = d/2$. Verified for $d \leq 7$ (Check V4).

Remark 4.4 ($d \leq 2$). For $d = 2$, $n = 1$: the only border length is $p = 0$ and the condition is $w_1 \neq w_1$, false. So there are no single-shape solutions at $d \leq 2$, consistent with [2, Thm. 6.1] (nonvanishing for $\gcd \leq 2$).

Theorem 4.5 (single-shape weights, strong condition). *In the situation of Theorem 4.2, $\tau = c\mathbb{K}_w$ satisfies $(2B_d^{**})$ if and only if w is an **antipalindrome**, $w_i \neq w_{n+1-i}$ for all i . In particular there is no such w when n is odd, i.e. when d is even; and for d odd there are exactly $2^{(d-1)/2}$ of them.*

Proof. Fix δ and γ and consider $\tau(\alpha\gamma\alpha')\tau(\beta\gamma\beta')$ with $|\alpha| = |\beta'| = \delta - 1$, $|\alpha'| = |\beta| = n - \delta$. It is nonzero iff $\alpha\gamma\alpha' = w$ and $\beta\gamma\beta' = w$. In the first word γ sits at position $|\alpha| + 1 = \delta$, in the second at position $|\beta| + 1 = n - \delta + 1$; the remaining blocks are then determined and unconstrained. So the γ -side is nonzero iff $\gamma = w_\delta = w_{n+1-\delta}$, with value c^2 . Hence the two sides of (2) agree iff both vanish, iff $w_\delta \neq w_{n+1-\delta}$. Ranging over $\delta \in [1, n]$ gives the antipalindrome condition. If n is odd, $\delta = (n + 1)/2$ demands $w_\delta \neq w_\delta$; if n is even the condition determines $w_{[n/2+1, n]}$ freely from $w_{[1, n/2]}$. $\square$

Remark 4.6. An antipalindrome is non-overlapping: if p is a border length then, comparing w_{p+1} with $w_{n-p} = w_{n+1-(p+1)}$, the antipalindrome condition at $i = p + 1$ gives $w_{p+1} \neq w_{n-p}$ directly. So Theorem 4.5 describes a subfamily of Theorem 4.2, as it must, and the inclusion is strict for $d \geq 4$ (e.g. $d = 4$: no antipalindromes, four non-overlapping words). This is one more instance of [2, Rem. 5.2]: $(2B^{**})$ is strictly stronger than $(2B^*)$ from $d \geq 4$ on.

5 Operators that add a single ribbon shape and commute

Theorem 5.1. *Let $d \geq 3$, let $w \in \{0, 1\}^{d-1}$ be non-overlapping, let $\alpha = \alpha(w) \models d$ be the corresponding ribbon shape, let K be any integral domain and let $k \geq 2$. Put $e = d$, $f = kd$ and*

$$W = \mathbb{K}_w \text{ (rank } d), \quad \bar{W}(y) = (-1)^{\text{ht}(y)} \prod_{i=1}^k [y_{[(i-1)d+1, id-1]} = w] \text{ (rank } kd).$$

Then:

- (i) $R_d^W s_\lambda = \sum_\mu s_\mu$, the sum over those d -ribbons μ/λ of shape exactly α ; the operator adds one ribbon shape and nothing else.
- (ii) $\bar{W}$ is supported on exactly 2^{k-1} of the 2^{kd-1} shapes of rank kd , namely on the words $w c_1 w c_2 \cdots c_{k-1} w$ with $c_i \in \{0, 1\}$ free.
- (iii) $[R_d^W, R_{kd}^{\bar{W}}] = 0$.

Since $\gcd(d, kd) = d \geq 3$, each such pair is a counterexample to [1, Lemma 8.2], and W has the smallest support any counterexample can have.

Proof. (i) is the definition of R_d^W together with [1, Lemma 1.3], and (ii) is the definition of $\tau^{\otimes k}$ ([2, Conv. 1.1]): $\sigma(u) = \prod_i \tau(u_{[(i-1)d+1, id-1]})$, the $k - 1$ letters in positions $d, 2d, \dots, (k - 1)d$ being ignored.

For (iii) put $\tau := \mathbb{K}_w$, so that $\sigma = \tau$ (as $e/d = 1$) and $\bar{\sigma} = \tau^{\otimes k}$, and note $W = (-1)^{\text{ht}(w)} \sigma$ pointwise; rescaling W by the unit $(-1)^{\text{ht}(w)}$ does not affect whether the commutator vanishes. By Theorem 4.2, τ satisfies $(2B_d^*)$, which for $e = d$ is $(2B_e^*)$ for σ . Now apply [2, Thm. 1.3], direction $(\Leftarrow)$, with $c = \bar{c} = 1$: the hypotheses are $\sigma = \tau^{\otimes(e/d)}$, $\bar{\sigma} = \tau^{\otimes(f/d)}$ and $(2B_e^*)$ for σ , all of which hold. Hence the commutator vanishes.

Finally, a weight with empty support is $\equiv 0$, which [1, Lemma 8.2] excludes by hypothesis; so a single point is the smallest possible support. $\square$

Corollary 5.2 (the smallest new case). *Let K be any integral domain. Let W be the rank-4 shape weight with*

$$W_{(1,3)} = -1, \quad W_\beta = 0 \text{ for the other seven compositions } \beta \models 4,$$

and let $\bar{W}$ be the rank-8 weight with

$$\bar{W}_{(1,4,3)} = +1, \quad \bar{W}_{(1,3,1,3)} = -1, \quad \bar{W}_\beta = 0 \text{ for the other 126 compositions } \beta \models 8.$$

Then $[R_4^W, R_8^{\bar{W}}] = 0$ and both operators are nonzero.

Proof. Theorem 5.1 with $d = 4$, $k = 2$, $w = (0, 0, 1)$: the border lengths of 001 are $p = 0$ only, and $w_1 = 0 \neq 1 = w_3$, so w is non-overlapping; $\alpha(001) = (1, 3)$. The support of $\tau^{\otimes 2}$ is $\{001c001 : c \in \{0, 1\}\} = \{0010001, 0011001\}$, with $\alpha(0010001) = (1, 4, 3)$ and $\alpha(0011001) = (1, 3, 1, 3)$, of heights 2 and 3, so $(-1)^{\text{ht}}$ is $+1$ and -1 respectively; and $\text{ht}(001) = 1$ gives $W_{(1,3)} = -1$. Verified directly in §9, V5. $\square$

Remark 5.3. The list of admissible single shapes begins:

$$\begin{aligned} d = 3 : & (1, 2), (2, 1) \\ d = 4 : & (1, 3), (3, 1), (1, 1, 2), (2, 1, 1) \\ d = 5 : & (1, 4), (4, 1), (1, 1, 3), (3, 1, 1), (1, 2, 2), (2, 2, 1), (1, 1, 1, 2), (2, 1, 1, 1) \end{aligned}$$

with 2, 4, 8, 12, 28, 48, 104 shapes for $d = 3, \dots, 9$. The list is stable under ribbon transposition, which by [1, Lemma 1.6] is reversal-and-complementation on words: both operations preserve border lengths and swap the two sides of (NB). At $d = 3$ it recovers [2, Thm. 6.7] ($w = (1, 0)$, shape $(2, 1)$) and its transpose.

6 The structure of an admissible support

We now drop the single-shape hypothesis. Recall from [2, Conv. 6.4], for $1 \leq j \leq d-1$ with $2j \neq d$ and $b \in \{0, 1\}$,

$$\mathcal{C}_{j,b} = \{w \in \{0, 1\}^n : w_j = b, w_{d-j} = 1-b\}, \quad |\mathcal{C}_{j,b}| = 2^{d-3}.$$

Note $\mathcal{C}_{j,b} = \mathcal{C}_{d-j,1-b}$, so there are $2\lfloor(d-1)/2\rfloor$ distinct cylinders.

Theorem 6.1. *Let $d \geq 2$ and let $Z \subseteq \{0, 1\}^n$ be admissible (Definition 2.2). Then either $Z = \emptyset$, or $Z = \{0, 1\}^n$, or there are $j \in [1, d-1]$ with $2j \neq d$ and $b \in \{0, 1\}$ with $Z \subseteq \mathcal{C}_{j,b}$. Moreover, in the third case let j_0 be minimal such; then Z is invariant under flipping any single coordinate in*

$$\{1, \dots, j_0 - 1\} \cup \{d - j_0 + 1, \dots, d - 1\}.$$

Proof. For $j \in [1, d-1]$ put $\delta := d - j$ and, for $\pi' \in \{0, 1\}^{j-1}$ and $b \in \{0, 1\}$,

$$A_b(\pi') = \{\pi \in \{0, 1\}^{d-j-1} : \pi b \pi' \in Z\}, \quad B_b(\pi') = \{\zeta \in \{0, 1\}^{d-j-1} : \pi' b \zeta \in Z\}.$$

The lengths match those in Definition 2.2: $|\pi| = |\zeta| = \delta - 1 = d - j - 1$ and $|\pi'| = d - \delta - 1 = j - 1$. In $\pi b \pi'$ the letter b sits at position $\delta = d - j$; in $\pi' b \zeta$ it sits at position $d - \delta = j$. Condition (3) at δ says precisely

$$A_0(\pi') \times B_0(\pi') = A_1(\pi') \times B_1(\pi') \quad \text{for every } \pi'. \quad (5)$$

Let $F_0 = \emptyset$ and $F_j = F_{j-1} \cup \{j, d-j\}$, so $F_j = \{1, \dots, j\} \cup \{d-j, \dots, d-1\}$. Write $P(j)$ for the statement “ Z is invariant under flipping any single coordinate in F_j ”; $P(0)$ is vacuously true.

Inductive step. Fix $j \geq 1$ and assume $P(j-1)$, i.e. invariance at every coordinate in $\{1, \dots, j-1\} \cup \{d-j+1, \dots, d-1\}$. In $\pi b \pi'$ the block π' occupies positions $d-j+1, \dots, d-1$, and in $\pi' b \varsigma$ it occupies positions $1, \dots, j-1$. By $P(j-1)$, membership in Z is unchanged by flipping any of these positions, so $A_b(\pi')$ and $B_b(\pi')$ do not depend on π' ; write A_b, B_b .

Two subsets $A_0 \times B_0$ and $A_1 \times B_1$ of a product are equal iff either both are empty or $A_0 = A_1$ and $B_0 = B_1$ (a nonempty product determines its factors as its projections). So (5) leaves exactly two cases.

Case 1: $A_0 = A_1$ and $B_0 = B_1$. $A_0 = A_1$ says $\pi 0 \pi' \in Z \iff \pi 1 \pi' \in Z$ for all π, π' , i.e. Z is invariant under flipping coordinate $\delta = d-j$; $B_0 = B_1$ says the same for coordinate j . Together with $P(j-1)$ this is $P(j)$.

Case 2: both products empty. Then $(A_0 = \emptyset \text{ or } B_0 = \emptyset)$ and $(A_1 = \emptyset \text{ or } B_1 = \emptyset)$. Now $A_b = \emptyset$ says no word of Z has b in position $d-j$, and $B_b = \emptyset$ says no word of Z has b in position j . Hence:

- $A_0 = A_1 = \emptyset$, or $B_0 = B_1 = \emptyset$: every word of Z would have to avoid both values at one position, so $Z = \emptyset$.
- $A_0 = \emptyset$ and $B_1 = \emptyset$: every $w \in Z$ has $w_{d-j} = 1$ and $w_j = 0$, i.e. $Z \subseteq \mathcal{C}_{j,0}$.
- $A_1 = \emptyset$ and $B_0 = \emptyset$: symmetrically $Z \subseteq \mathcal{C}_{j,1}$.

If $2j = d$ the last two read $Z \subseteq \{w : w_j = 0, w_j = 1\} = \emptyset$, consistent with $\mathcal{C}_{j,b}$ being undefined (empty) there.

Termination. Let $J = \lceil (d-1)/2 \rceil$; then $J \leq d-1$ for $d \geq 2$, and $F_J = \{1, \dots, J\} \cup \{d-J, \dots, d-1\} = \{1, \dots, d-1\}$ because the gap $\{J+1, \dots, d-J-1\}$ is empty when $2J \geq d-1$. Run the inductive step for $j = 1, 2, \dots, J$. If Case 2 never occurs then $P(J)$ holds, so Z is invariant under flipping every coordinate, hence $Z \in \{\emptyset, \{0,1\}^n\}$. Otherwise let j_1 be the first index at which Case 2 occurs; then $Z = \emptyset$ or $Z \subseteq \mathcal{C}_{j_1,b}$ for some b .

Minimality. Suppose $Z \neq \emptyset$ and $Z \subseteq \mathcal{C}_{j_1,b}$. For $j < j_1$ Case 1 held, so Z is invariant under flipping coordinate j ; but $Z \subseteq \mathcal{C}_{j,b'}$ would force $w_j = b'$ for all $w \in Z$, incompatible with that invariance unless $Z = \emptyset$. Hence no $j < j_1$ has $Z \subseteq \mathcal{C}_{j,b'}$, so $j_0 = j_1$, and $P(j_0 - 1)$ is exactly the asserted invariance. $\square$

Corollary 6.2 (why the threshold is 3). *For $d \leq 2$ the only admissible sets are $\emptyset$ and $\{0,1\}^n$; a rank- d weight satisfying $(2B_d^*)$ and $\neq 0$ therefore vanishes nowhere. For $d \geq 3$ proper nonempty admissible sets exist, e.g. $\mathcal{C}_{1,b}$.*

Proof. For $d = 1$ there is nothing to prove ($n = 0$). For $d = 2$ the only index is $j = 1 = d/2$, so Theorem 6.1 leaves only $\emptyset$ and $\{0,1\}^1$; combined with Proposition 2.3 this gives the claim. For $d \geq 3$ the index $j = 1$ satisfies $2j \neq d$, and $\mathcal{C}_{1,b}$ is admissible by [2, Lemma 6.5]. $\square$

This is the conceptual content of the threshold “ $\gcd(e, f) \geq 3$ ” in [2, Thm. 1.4]: $d \geq 3$ is exactly the condition that the index set $[1, d-1]$ contains some j with $j \neq d-j$. Corollary 6.2 reproves the support half of [2, Thm. 6.1] (nonvanishing for $\gcd \leq 2$) without any of its case analysis.

Corollary 6.3 (the maximal proper supports are exactly the cylinders). *Let $d \geq 3$. The maximal proper admissible subsets of $\{0,1\}^n$ are precisely the $2\lfloor (d-1)/2 \rfloor$ distinct cylinders $\mathcal{C}_{j,b}$.*

Proof. Each $\mathcal{C}_{j,b}$ is admissible [2, Lemma 6.5] and proper, having $2^{d-3} < 2^{d-1}$ elements. By Theorem 6.1 every proper admissible set is contained in some cylinder, so every maximal one is a

cylinder. Distinct cylinders all have the same cardinality 2^{d-3} , so none strictly contains another, and each is therefore maximal. Verified for $d \leq 7$ (Check V3). $\square$

Corollary 6.4 (the quarter dichotomy). *Let $\tau \not\equiv 0$ be a rank- d shape weight over an integral domain satisfying $(2B_d^*)$. Then either τ vanishes nowhere, or*

$$\#\{u : \tau(u) \neq 0\} \leq 2^{d-3} = \frac{1}{4} \cdot 2^{d-1},$$

so τ vanishes at more than three quarters of all ribbon shapes of rank d . There is nothing in between.

Consequently, if $[R_e^W, R_f^{\bar{W}}] = 0$ with $W \not\equiv 0 \not\equiv \bar{W}$ and $d = \gcd(e, f)$, then either W is nowhere zero, or W is supported on at most a fraction $4^{-e/d}$ of the 2^{e-1} ribbon shapes of rank e — and likewise for $\bar{W}$ with f in place of e .

Proof. The first statement is Proposition 2.3 and Corollary 6.3: the support is admissible, hence empty (excluded), everything, or inside a cylinder, which has 2^{d-3} elements. For the second, [2, Thm. 1.3] gives $\sigma = c\tau^{\otimes k}$ with $k = e/d$, and $|\text{supp } \tau^{\otimes k}| = |\text{supp } \tau|^k \cdot 2^{k-1}$, the $k-1$ ignored letters being free. If τ is nowhere zero this is $2^{k(d-1)+k-1} = 2^{e-1}$; otherwise it is at most $2^{k(d-3)+k-1} = 2^{e-1-2k}$. $\square$

This is the shape the statement should have taken from the start: not “no value vanishes” but “either none vanishes or almost all do”. The refuted [1, Lemma 8.2] asserted the first alternative of a genuine dichotomy and lost the second.

Remark 6.5 (admissibility is a Horn condition). Theorem 7.2 exhibits (3) as a conjunction of definite Horn clauses $x_u \wedge x_v \rightarrow x_{u'}$ in the 2^n propositional variables $x_w = [w \in Z]$, and Proposition 7.3 is the classical fact that a class of subsets is Horn-definable iff it is closed under intersection. This is, I think, the right expectation to have about (H2a) below: counting the models of a Horn theory is not the sort of thing that has a product formula, and the counts 1, 3, 11, 31, 356 of §10 do not look like one.

Remark 6.6 (the symmetry group). Admissibility is preserved by complementing all letters ($w \mapsto \bar{w}$) and by reversal ($w \mapsto w^{\text{rev}}$). Complementation exchanges the two sides of (3); reversal carries the relation at δ to the relation at $d - \delta$ with the roles of the two factors exchanged, the shared block π' going to its reverse. The composite is ribbon transposition ([1, Lemma 1.6]). Accordingly $\mathcal{C}_{j,b}$ is carried to $\mathcal{C}_{j,1-b}$ and $\mathcal{C}_{d-j,b} = \mathcal{C}_{j,1-b}$ respectively.

7 Generation: the admissible sets are a closure system

Theorem 6.1 says where an admissible set can live. This section says how to build one. Both statements are short; together they are what I can offer in place of the closed classification that (H2) asks for.

Definition 7.1. Let $u, v \in \{0, 1\}^n$ and $\delta \in [1, n]$. The pair (u, v) is δ -linked if

$$u_{[\delta+1, n]} = v_{[1, n-\delta]} \quad \text{and} \quad u_\delta = v_{n-\delta+1}.$$

For such a pair write $\iota_\delta(u, v) := (u^{(\delta)}, v^{(n-\delta+1)})$, where $x^{(i)}$ is x with the letter in position i flipped.

Flipping position δ of u does not touch $u_{[\delta+1, n]}$ and flipping position $n - \delta + 1$ of v does not touch $v_{[1, n-\delta]}$, so ι_δ carries δ -linked pairs to δ -linked pairs, and it is an involution.

Theorem 7.2 (generation). $Z \subseteq \{0,1\}^n$ is admissible if and only if it is closed under the rule

$$u, v \in Z \text{ } \delta\text{-linked} \implies u^{(\delta)} \in Z \text{ and } v^{(n-\delta+1)} \in Z. \quad (\text{R})$$

Proof. Fix δ and a triple (π, π', ς) as in Definition 2.2, and put $u_\gamma = \pi\gamma\pi'$, $v_\gamma = \pi'\gamma\varsigma$. Both (u_0, v_0) and (u_1, v_1) are δ -linked, and $\iota_\delta(u_0, v_0) = (u_1, v_1)$. Conversely a δ -linked pair (u, v) comes from exactly one triple, namely $\pi = u_{[1, \delta-1]}$, $\pi' = u_{[\delta+1, n]} = v_{[1, n-\delta]}$, $\varsigma = v_{[n-\delta+2, n]}$. So, writing $P(u, v)$ for “ $u \in Z$ and $v \in Z$ ”, condition (3) at $(\delta, \pi, \pi', \varsigma)$ is exactly $P(u_0, v_0) \iff P(\iota_\delta(u_0, v_0))$, and (R) is the implication $P(u, v) \implies P(\iota_\delta(u, v))$ for every δ -linked pair. Since ι_δ is an involution on δ -linked pairs, the implication for all pairs is equivalent to the biconditional for all pairs. $\square$

Proposition 7.3. *The admissible subsets of $\{0,1\}^n$ are closed under arbitrary intersections, and $\{0,1\}^n$ is admissible. They therefore form a closure system: there is a closure operator $X \mapsto \langle X \rangle$, the least admissible set containing X , and the admissible sets are exactly the sets of the form $\langle X \rangle$.*

Proof. Let $(Z_t)_{t \in T}$ be admissible and $Z = \bigcap_t Z_t$. For any u, v , “ $u \in Z$ and $v \in Z$ ” is the conjunction over t of “ $u \in Z_t$ and $v \in Z_t$ ”. So with P_t, P as in the previous proof, $P = \bigwedge_t P_t$ on both sides of (3), and a conjunction of biconditionals is a biconditional between the conjunctions. Admissibility of $\{0,1\}^n$ is immediate (both sides of (3) are always true). The rest is the standard fact about Moore families. $\square$

Remark 7.4. Theorems 4.2 and 7.2 now read as one statement: $\langle \{w\} \rangle = \{w\}$ if and only if w is non-overlapping. Indeed (R) applied to $u = v = w$ is nontrivial exactly when (w, w) is δ -linked, i.e. when w has period δ and $w_\delta = w_{n-\delta+1}$ — the failure condition of Theorem 4.2. The closure of a single word is computed in Check V7; its size is

$$\begin{array}{l|l} d=3 & 1 (\times 2), \quad 4 (\times 2) \\ d=4 & 1 (\times 4), \quad 8 (\times 4) \\ d=5 & 1 (\times 8), \quad 4 (\times 4), \quad 16 (\times 4) \\ d=6 & 1 (\times 12), \quad 2 (\times 4), \quad 8 (\times 8), \quad 32 (\times 8) \end{array}$$

— so a single word generates itself, or a cylinder (4 at $d=5$, 8 at $d=6$), or something smaller, or everything. Union is *not* an admissible operation: at $d=5$ both $\{0001\}$ and $\{0101\}$ are admissible and their union is not.

Remark 7.5 (where the ribbon structure enters). It is worth naming the step, because the analogous statement is false in general. A multiplicative structure on $\{0,1\}^n$ does *not* force its support to be a product set without a positivity hypothesis — this is why Hammersley–Clifford assumes $P > 0$ — so any argument here that used only “connectivity of the cube” would prove something known to be false. The proof of Theorem 6.1 uses no connectivity. What it uses is that the two-bead relation (1) couples position δ of one factor with position $m - \delta$ of the *other*, across a *shared block* π' . That is the ribbon input: it is [1, Lemma 4.3], where π' is the overlap of the two bead windows. The shared block is what makes $A_b(\pi')$ and $B_b(\pi')$ live on a common index set, so that (5) is an equality of *product sets* — and product-set rigidity, not connectivity, is what forces the dichotomy. The only way out is an asymmetry between positions j and $d - j$, which is exactly what $\mathcal{C}_{j,b}$ is.

8 Values on a given support

Fix an admissible Z and let F be a field. By Proposition 2.3 the solutions of $(2B_d^*)$ with support exactly Z form a torsor under the group

$$\text{Hom}(\mathbb{Z}^Z / L_Z, F^\times), \quad L_Z = \langle e_{u_0} + e_{v_0} - e_{u_1} - e_{v_1} : u_0, v_0, u_1, v_1 \in Z \rangle,$$

the generators running over the relations (1) *all four of whose arguments lie in Z* (the others are $0 = 0$), with base point $\mathbb{K}_Z$. The invariant factors of L_Z were computed for all admissible Z at $d \leq 6$ (Check V6). Sample:

$$\begin{array}{llll} d=3 & |Z|=1 : (F^\times)^1 & |Z|=4 : (F^\times)^2 & \\ d=4 & |Z|=1 : (F^\times)^1 & |Z|=2 : (F^\times)^2 & |Z|=8 : (F^\times)^2 \times \mu_2 \\ d=6 & |Z|=1 : (F^\times)^1 & |Z|=3 : (F^\times)^3 & |Z|=32 : (F^\times)^3 \times \mu_2 \end{array}$$

where $\mu_2 = \{x \in F^\times : x^2 = 1\}$. Two remarks. First, a single-shape support always gives a one-dimensional torus, the free scalar c of Theorem 4.2 — as it must, there being no relations with all four arguments in a singleton unless $\pi 0 \pi' = \pi 1 \pi'$, which is impossible. Second, 2-torsion does occur (already at $d = 4$ on the full support), so the solution set is not a torus over every field and in characteristic 2 some sign choices collapse; this matches [1, Cor. 5.4], whose group has a μ_2 factor.

9 Verification

All code is in `proofs/code/2026-09-16-c2-Q150-H2/`. The operator engine (`independent_maya.py`) was written from [1, Conv. 1.2] alone and shares no code with the Q149 or Q150 engines; it acts on Maya sets directly, so a state need not be given as a small partition.

V1 (calibration of the engine). [1, Lemma 1.3] re-derived against Young diagrams: 1375 bead moves ($|\lambda| \leq 8$, $e \leq 5$), 0 mismatches.

V2 ([2, Thm. 6.7], and controls). $[R_3^W, R_6^{\bar{W}}]$ vanishes on all 195 partitions with $|\lambda| \leq 11$. Negative controls over the 97 partitions with $|\lambda| \leq 9$, all of which fire: NC1 $W_{(1,2)}=1$, 35 nonzero; NC2 $W_{(3)}=1$, 32; NC3 $W_{(1,1,1)}=1$, 32; NC4 sign flip on $\bar{W}_{(2,1,2,1)}$, 2; NC5 drop $\bar{W}_{(2,1,2,1)}$, 2; NC6 drop $\bar{W}_{(2,3,1)}$, 2; NC7 add $\bar{W}_{(6)}=1$, 48. Positive control: Murnaghan–Nakayama ($\sigma \equiv 1$, $\bar{\sigma} \equiv 1$) gives 0 nonzero, as it must. NC1 is the control that located the omission of §3.

V3 (admissible supports). All $Z \subseteq \{0,1\}^n$ satisfying (3) were enumerated for $d \leq 7$ by constraint propagation, and independently by brute force over all 2^{2^n} subsets for $d \leq 5$. The two agree: 4, 8, 26 admissible sets at $d = 3, 4, 5$. (The propagation search was wrong on its first run — it failed to undo assignments made during a propagation that ended in a conflict — and the brute-force comparison at $d = 4$, 2 versus 8, is what caught it.) Totals, and proper nonempty counts:

d	3	4	5	6	7
admissible	4	8	26	66	722
proper $\neq \emptyset$	2	6	24	64	720

For each of these, Theorem 6.1 was checked directly: every proper nonempty Z lies in some $\mathcal{C}_{j,b}$ (720/720 at $d = 7$), and for the minimal such j_0 the asserted flip-invariances hold in every case. The distribution of j_0 at $d = 7$ is 712 at $j_0 = 1$, 6 at $j_0 = 2$, 2 at $j_0 = 3$; the j_0 values factor as: $2 \cdot 2^{d-3}$ sets of size 2^{d-3} are the cylinders themselves, one for each of the $2 \lfloor (d-1)/2 \rfloor$ distinct cylinders, and at $d = 7$ that is 6 sets of size 16 — matching the 6 distinct $\mathcal{C}_{j,b}$. Also checked: $\mathbb{K}_Z$ solves (1) for every admissible Z (Proposition 2.3), all $d \leq 7$.

V4 (Theorems 4.2 and 4.5). For every $d \leq 9$ and every $w \in \{0, 1\}^{d-1}$ the criteria (NB) and “antipalindrome” were compared against a direct evaluation of (1), resp. (2), over all instances. Exact agreement, both conditions, all d . Counts of non-overlapping w : 0, 2, 4, 8, 12, 28, 48, 104 for $d = 2, \dots, 9$; of antipalindromes: 0, 2, 0, 4, 0, 8, 0, 16. Note the counts are 2^{d-2} (the $p = 0$ clause alone) exactly for $d \leq 5$ and strictly smaller from $d = 6$ on, which is the content of Remark 4.3.

V5 (Theorem 5.1 end to end, with a liveness check). A commutator that vanishes because its two-bead sector is never reached is no evidence, so each scan reports how many states actually reach that sector. Exhaustive over all 2^{13} (resp. 2^{15}) Maya windows: at $(e, f) = (3, 6)$, $w \in \{01, 10\}$, 0 nonzero commutators with 187 resp. 508 states reaching the two-bead sector; at $(4, 8)$, $w \in \{001, 011, 100, 110\}$, 0 nonzero with 71, 71, 514, 200 states reaching it. Corollary 5.2 is the case $w = 001$. Randomised scans over 3000 Maya states each also give 0 for every non-overlapping w at $(d, k) \in \{(3, 2), (4, 2), (5, 2), (6, 2), (7, 2), (5, 3)\}$.

V5' (the negative control for Theorem 5.1, constructed not sampled). Random scans are the wrong instrument here: at $d = 7$ the witnessing window has width 21 and 56 000 random samples at seven densities found nothing, which would have read as a silent control. Instead the witness is built from the geometry of the two-bead element [1, Lemma 4.3]: bead A at site 0 moving $0 \mapsto d$, bead B at site δ moving $\delta \mapsto \delta + kd$, with the window filled so that the σ -word is w and the $\bar{\sigma}$ -word is $w0w \cdots 0w$. For every w with $w_1 \neq w_{d-1}$ that fails (NB) — the hard cases, the ones the $p = 0$ clause does not catch — and for every $d \leq 10$, the constructed state has a nonzero two-bead commutator element: 112/112 fire. The same construction applied to non-overlapping w gives zero at every δ , for all $d \leq 7$. (An earlier version of this check used $\delta = n + 1 - p$ instead of $\delta = n - p$ and reported zero everywhere; the fact that the *positive* cases also reported zero is what exposed it.)

V7 (generation). Theorem 7.2 was checked against a direct evaluation of (3): over all 2^{2^n} subsets for $d \leq 4$, and over the admissible sets together with all 2-element sets for $d = 5, 6$. No mismatches. Intersection closure (Proposition 7.3) was checked on every pair of admissible sets for $d \leq 6$. The closure of every single word was computed for $d \leq 6$; sizes as tabulated in §7.

V6 (values). For every admissible Z at $d \leq 6$ the lattice L_Z of §8 was put in Smith normal form. Free ranks and torsion as tabulated there; 2-torsion occurs at $d = 4$ ($|Z| = 8$), $d = 6$ ($|Z| \in \{2, 5, 8, 32\}$) and nowhere else in range.

Scope, against the quantifier. Theorems 4.2, 4.5, 6.1 and 5.1 are proved for all d ; the computations above are confirmations, not the proof. The quantifier ranges checked are: all $w \in \{0, 1\}^{d-1}$ for $d \leq 9$ (V4); all $Z \subseteq \{0, 1\}^{d-1}$ for $d \leq 7$ (V3); all Maya windows of width 13, 15 for the two smallest (e, f) (V5); all hard failing w for $d \leq 10$ (V5'). Nothing is checked for $k \geq 4$, and nothing for $e > d$ — the latter is exactly gap (H1) of [2], untouched here.

10 What is and is not proved

Proved

1. Proposition 2.3: admissible supports = supports of solutions, with $\mathbb{K}_Z$ as base point.

2. Theorem 4.2: $c \not\models_w$ satisfies $(2B_d^*)$ iff w is non-overlapping. Unconditional, one domain appeal, witness c .
3. Theorem 4.5: $c \not\models_w$ satisfies $(2B_d^{**})$ iff w is an antipalindrome; none for even d .
4. Theorem 5.1 and Corollary 5.2: for every $d \geq 3$, every non-overlapping w and every $k \geq 2$, an operator adding the single ribbon shape $\alpha(w)$ commutes with an explicit rank- kd operator supported on 2^{k-1} shapes.
5. Theorem 6.1: the containment half of (H2), with the flip-invariance refinement; and Corollary 6.2, which recovers the $\gcd \leq 2$ nonvanishing of [2, Thm. 6.1] at the level of supports and explains the threshold.
6. Corollaries 6.3 and 6.4: the cylinders are exactly the maximal proper supports, whence the quarter dichotomy — a nonzero commuting weight is nowhere zero or supported on at most a quarter (rank d), resp. $4^{-e/d}$ (rank e), of the shapes.
7. Theorem 7.2 and Proposition 7.3: Z is admissible iff closed under the linked-pair flip rule (R); the admissible sets form a closure system, and Theorem 4.2 is the statement $\langle \{w\} \rangle = \{w\}$.
8. §3: independent re-verification of [2, Thm. 6.7] and the correction $W_{(1,2)} = 0$ to its statement.

Not proved

- (H2a) **Which subsets of a cylinder are admissible.** Theorem 6.1 reduces the classification to: for each j with $2j \neq d$, describe the admissible $Z \subseteq \mathcal{C}_{j,b}$ that are flip-invariant at $\{1, \dots, j-1\} \cup \{d-j+1, \dots, d-1\}$. Such a Z is the same thing as a set $S \subseteq \{0, 1\}^{d-2j-1}$ of *middle* words, and unwinding (3) for δ in the remaining range turns the condition into: for $1 \leq i \leq m-1$ ($m = d-2j-1$) and all $p, r \in \{0, 1\}^{i-1}$, $q \in \{0, 1\}^{m-i-1}$,

$$[(p, c, b, q) \in S \wedge (q, 1-b, c, r) \in S] \text{ is the same for } c = 0 \text{ and } c = 1.$$

This is (3) again but with a pinned letter adjacent to the toggled one on each side and a shared block one shorter — structurally similar, not literally the same, so it does not self-reduce. I verified the reduction at $j = 1$, $d \leq 6$ but do not have a closed description of its solutions. The counts of admissible S are 1, 3, 11, 31, 356 for $m = 0, \dots, 4$, which I do not recognise; the jump $31 \rightarrow 356$ suggests the answer is not a product formula. *Theorem 4.2 is the case $|S| = 1$ of this question, solved.* By Theorem 7.2 and Proposition 7.3 the question has been turned into a concrete one about an explicit closure operator — “describe the closed sets of (R)” — rather than about a system of equations; that is progress of a kind, but it is not the answer.

- (H2b) **A uniform description of the value lattice L_Z .** §8 computes it case by case; I have no formula for its rank in terms of Z , and the sporadic 2-torsion is unexplained.
- (H1) **Untouched.** The middle condition of [2, Rem. 5.2] — which τ have $\tau^{\otimes k}$ satisfying $(2B_{kd}^*)$ — is not addressed. Note however that Theorem 4.5 gives the first clean statement about it from the $(2B^{**})$ side: for even d no single-shape τ satisfies $(2B_d^{**})$, yet by Theorem 5.1 single-shape τ at $d = 4, 6, 8$ do produce commuting pairs at $e = d$. So for even d the route through $(2B^{**})$ of [2, Thm. 6.6] misses the single-shape examples entirely — which is why they are new.

Effect on [2]

No theorem of [2] changes. Theorem 6.1 upgrades its computational observation (“the admissible supports at $d \leq 5$ are the full support and the subsets of the $\mathcal{C}_{\delta,b}$ ”, §8 (H2)) to a theorem for all d , and Corollary 6.2 gives an independent proof of the support half of its Theorem 6.1. Theorem 5.1 enlarges the family of counterexamples in its Theorem 6.6 — strictly, since for even d that theorem produces none of them — and drives the support down to a single point, which is the extreme case. The only textual change required is the omitted value in its Theorem 6.7, §3.

References

- [1] Clio, *Commuting ribbon operators with shape-dependent weights*, 2026-09-15, `proofs/2026-09-15-c1-Q149-shape-weights.tex`.
- [2] Clio, *The nonvanishing lemma for ribbon shape weights is false, and the true statement is a commuting-words theorem*, 2026-09-16 (cycle 1), `proofs/2026-09-16-c1-Q150-nonvanishing.tex`.